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Electric Eels Inspired Iontronic Artificial Skin with Multimodal Perception and In-Sensor Reservoir Computing

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As the largest sensory organ, the human skin generates ionic signals in response to tactile, thermal, and electrical stimuli, which are then transmitted to neurons and processed by brain, thereby enabling sensing and memory, ultimately promoting conscious perception and decision-making. However, existing artificial skins face significant challenges including the inability to achieve multimodal perception and memory simultaneously (i.e., tactile, thermal, and electrical stimuli), difficulty in detecting ultra-low currents, and limitations in rich synaptic behaviors that are essential for highly efficient in-sensor reservoir computing. Inspired by electric eels, the study here develops an artificial skin based on iontronic p-n junctions consisting of PolyAT and PolyES bi-layered structures. This skin features broad detection ranges for temperature (−80 to 120 °C, well beyond the reach of hydrogel counterparties), pressure (0.075 Pa to 400 kPa, among the highest sensitivities ever reported), and current (1–200 nA), meanwhile demonstrates rich synaptic behaviors and memory functions. Additionally, incorporating the iontronic skin in a robotic hand can grasp objects with different temperatures and weights on demand. Further, a fully memristive in-sensor reservoir computing is implemented on the iontronic skin, allowing sensing, decoding, and learning via electrical stimulation, achieving 91.3% accuracy in classifying MNIST handwritten digit images.

1. Introduction

The human skin is the largest sensory organ, renowned for its exceptional stretchability, moisture retention, and freeze resistance.[1–3] It generates ionic signals in response to tactile, thermal, and electrical stimuli, enabling sensing, transmission, processing, and memory, thereby promoting conscious perception and decision-making.[4,5] Inspired by these features, bioinspired ionic flexible sensors have garnered significant attention over past decades. These sensors utilize ions for signaling and replicate skin-like properties such as multimodal perception, memory, and synaptic behaviors.[6–11] Compared to traditional ionic gels, polyionic liquid (PIL) elastomers exhibit significant differences in structure and physical properties. Traditional ionic gels consist of ionic conductors and polymer matrices, typically forming a semi-solid gel structure that relies on solvents (usually water) to provide ionic conductivity and maintain mechanical performance. While this structure is widely used in electrochemical devices, the evaporation or freezing of water can lead to performance degradation and may also cause ionic leakage. In contrast, PIL elastomers are entirely composed of a cross-linked polyelectrolyte network and associated ion pairs, without any liquid components. Due to the low volatility and excellent thermal stability of ionic liquids, PIL elastomers demonstrate high elasticity, superior mechanical strength, and good conductivity at room temperature.[8,12–16] Unlike traditional ionic gels, these materials are not affected by issues such as water evaporation or freezing, making them more stable in various

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environments, particularly in applications requiring long-term, stable performance.[17–26] With their exceptional properties and unique advantages, PIL elastomers are becoming an important candidate in the next generation of functional materials, showing enormous potential in fields such as smart sensors, flexible electronics, and artificial skin. Despite these promises, reported iontronic skins still face several challenges: i) realizing multimodal perception and memory to tactile, thermal, and electrical stimuli, ii) detecting ultra-low currents at the nanoampere (nA) level, and iii) achieving rich synaptic behaviors essential for constructing in-sensor reservoir computing (RC).[27–30]

Interestingly, some organisms in nature, such as electric eels, use electrical signals to perform critical physiological functions.[31,32] The electric organ of an electric eel consists of thousands of electrocytes, which exhibit minimal voltage difference between the inside and outside of the cells during resting states but generate a significant potential difference when stimulated. Specifically, electrical signals are triggered within the electrocytes, activating ion channels to open and close, allowing ions to pass through the cell membrane and create a potential gradient across it.[33–35] The structure of electrocytes resembles a biological "diode". By controlling the discharge sequence of these cells, the electric eel transmits electrical signals to the brain, enabling it to carry out activities such as hunting, communication, and defense.

Inspired by the electric eel's skin, we developed a flexible, transparent, ion-leakage-free,[36,37] and biocompatible artificial skin based on iontronic p-n junction consisting of Poly(1-[2-acryloyloxyethyl]-3-butylimidazolium bis(trifluoromethane) sulfonimide) (PolyAT) and poly(1-ethyl-3-methyl imidazolium (3-sulfopropyl) acrylate) (PolyES) bi-layered structures (Figure 2a). In these elastomers, the anions or cations are fixed within the elastomer network, while counterions remain mobile. The polyanion/polycation heterojunction forms an ionic double layer (IDL), analogous to the depletion layer of a p-n semiconductor junction (**Figures 1f** and 2a; Figure S1, Supporting Information). Benefiting from the built-in potential, multifunctional artificial skin that demonstrates sensing and memory toward ultra-low electrical stimuli (≈1 nA), a wide pressure range (0.075 Pa to 400 kPa), and a broad temperature range (−80 to 120 °C well beyond the reach of hydrogel-based artificial skins) could be achieved simultaneously for the first time. In addition, it exhibits rich synaptic behaviors, including excitatory postsynaptic current (EPSC), excitatory postsynaptic potential (EPSP), paired-pulse facilitation (PPF), short-term plasticity (STP), long-term plasticity (LTP), short-term memory (STM), long-term memory (LTM), as well as multi-bit potentiation and depression. Leveraging these advantages, we have developed a multimodal artificial skin with an 8×8 sensor array capable of mapping temperature and pressure, and grasping of objects with specific temperatures and weights on demand when integrated with a robotic hand. Moreover, a fully memristive in-sensor RC system can be realized, which integrates the functions of transducers, message decoders, and artificial intelligence (AI) for intelligent edge signal processing. The results demonstrate that the system can learn to classify representative Modified National Institute of Standards and Technology (MNIST) handwritten digit images transmitted via ion signal, achieving an accuracy of ≈91.3%. This work represents a remarkable advance in fully mimicking the functions of skin, and provides a hardware solution for intelligent artificial skins with multidimensional perception and memory functions, holding great potential for applications in wearable electronics, soft robotics, smart prosthetics, and augmented reality.

2. Results and Discussion

2.1. Design and Characterization of the PIL Elastomer Diode-Based Artificial Skin

The synthesis of 1-[2-acryloyloxyethyl]-3-butylimidazolium bis(trifluoromethane) sulfonimide imide (AT) and 1-ethyl-3-methyl imidazolium (3-sulfopropyl) acrylate (ES) monomers was carried out according to previous literature,[38] and the detailed synthesis steps are included in the Supporting Information. The AT and ES monomers were then mixed with the initiator and crosslinker in specific proportions, placed in a mold, and reacted overnight at 60 °C under a nitrogen atmosphere to prepare the elastomers (Figure S4, Supporting Information). Through the analysis of Fourier-transform infrared (FTIR) spectra (Figure S5, Supporting Information), the synthesis of PolyAT and PolyES was confirmed. The FTIR spectrum of PolyES shows a band at 3428 cm^{-1}, corresponding to the O-H stretching vibration, indicating that PolyES still contains trace amounts of water. The lightly crosslinked PIL elastomers exhibit excellent stretching performance. When PolyES and PolyAT were laminated into a layered structure, the mobile counterions near the interface diffused into the opposite regions driven by entropy. As a result, excess fixed polyanionic and polycationic charges remained on the PolyES and PolyAT sides, respectively (**Figure 2a**; Figure S1, Supporting Information). Since the crosslinked network restricts the long-range movement of fixed ions, an electric field forms across the interface, pointing from PolyAT to PolyES, where the drift current of mobile ions balances their diffusion current at equilibrium, giving rise to a built-in potential across the interface (Figure S1, Supporting Information).[39] The low glass transition temperature (Tg) of these PIL elastomers enables selective ion conduction at ambient temperatures through the segmental motion of polymer chains (Figure 2b).[40] Thermogravimetric analysis (TGA) revealed 5% weight loss temperatures of PolyAT and PolyES are ≈288 and 378 °C, respectively, indicating excellent thermal stability (Figure 2c). Figure 2d shows the stress–strain curves of PolyAT and PolyES, respectively. It can be seen that the elastic moduli of PolyAT and PolyES are 150 and 100 kPa, respectively. Meeting the mechanical performance requirements for flexible and wearable devices.

The PIL elastomer diode (Figure S6, Supporting Information) demonstrated good elasticity at sub-zero temperatures and could be reversibly twisted or bent without rupture, even at −20 °C (Figure S7, Supporting Information). In addition, the solvent retention of the PIL elastomer is significantly better than that of the hydrogel when stored at room temperature. After 30 days of room temperature storage, the PIL elastomer only experienced a slight mass loss (<3%), whereas the hydrogel lost almost all of its water after just two days (70% total mass loss) (Figure 2e). Furthermore, the PIL elastomer diode maintains excellent transparency at both room temperature and low temperature (−20 °C). However, the hydrogel becomes opaque when exposed to −20 °C (Figure S8, Supporting Information). These results confirm that

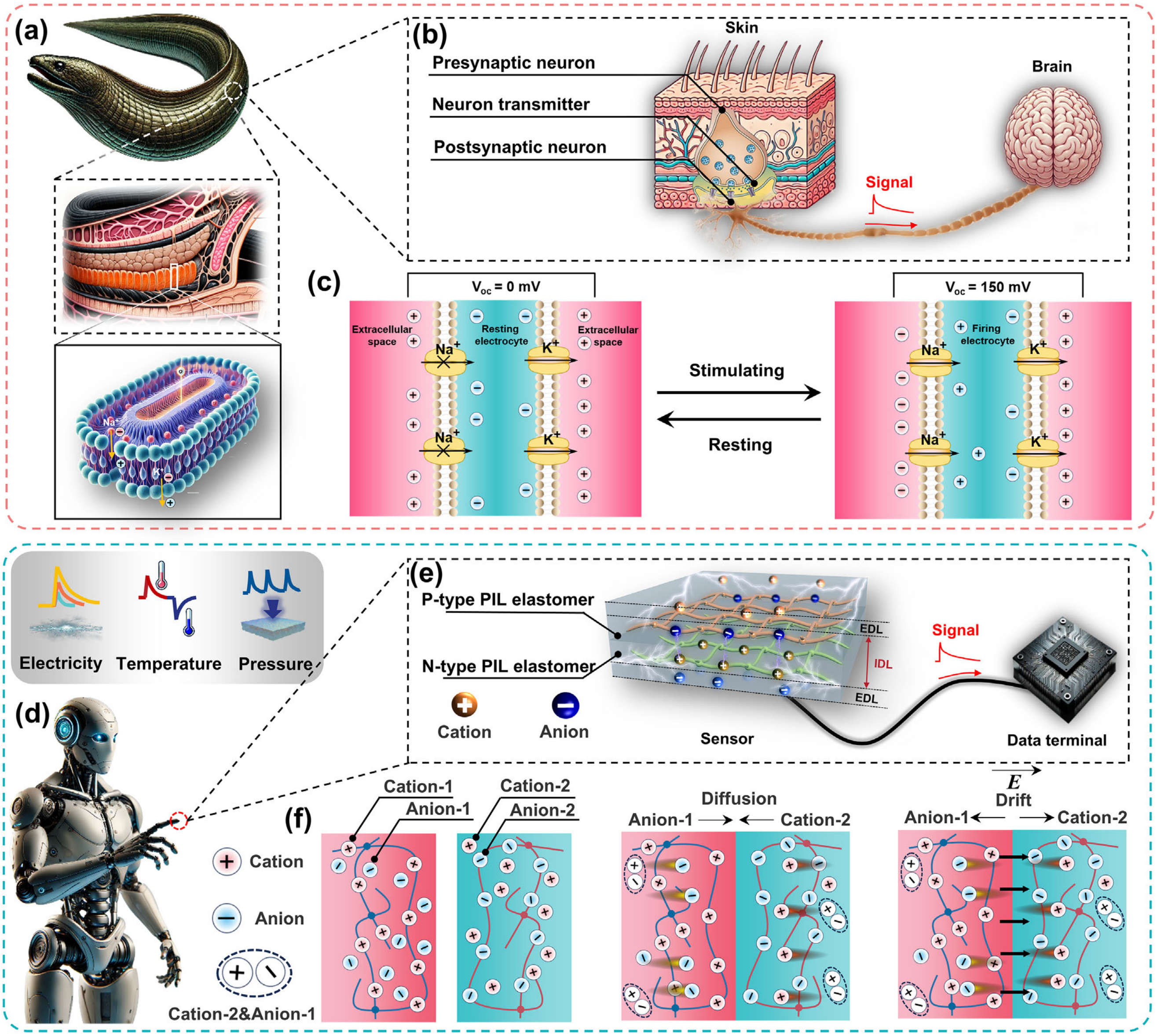

Figure 1. a) Electric eels. The first inset below shows the arrangement of electrocytes within the electric organs of electrophorus. The second inset below shows ion fluxes in the firing state. b) Structure of electric receptor in electrophorus. c) Mechanism of voltage generation in electrocytes. d) Schematic diagram of bio-inspired artificial skin integrated on robot fingers for multimodal perception and memory. e) The structure of artificial skin. f) PIL elastomer form a heterojunction by balancing two concurrent processes.

the PIL elastomer diode retains its functionality even under harsh conditions. Figure S9 (Supporting Information) illustrates the PIL elastomer can easily adhere to various substrates, including PET, glass, rubber, metal, and wood. To quantify the adhesion performance of PolyAT and PolyES on different substrates, we used the standard 90° peel test method to systematically measure their adhesion strength on various substrates. As shown in Figure 2f, the experimental results indicate that both PolyAT and PolyES exhibit excellent adhesion performance, forming strong interfacial bonds with various substrates (Figure S10, Supporting Information). Due to the depletion of mobile ions and the strong electrostatic adhesion provided by the fixed ions, a tight and seamless interface is formed between PolyAT and PolyES, which is visually confirmed by scanning electron microscopy (SEM) (Figure 2g). To further evaluate the stability of the interface, we subjected the sensor to rigorous environmental tests, including 100 cycles of high-temperature testing at 120 °C, 5 days of exposure to −20 °C, 100 stretching cycles, and 100 bending recovery cycles. The experimental results show that after undergoing these tests, the interface between PolyAT and PolyES remains intact and tight, with no noticeable delamination, demonstrating excellent morphological stability (Figure S11, Supporting Information). In addition, the interfacial stability between PolyAT, PolyES, and the PI-Cu electrodes in different environments has also been demonstrated (Figure S12, Supporting Information). Biocompatibility is crucial for artificial skin. We conducted a systematic evaluation

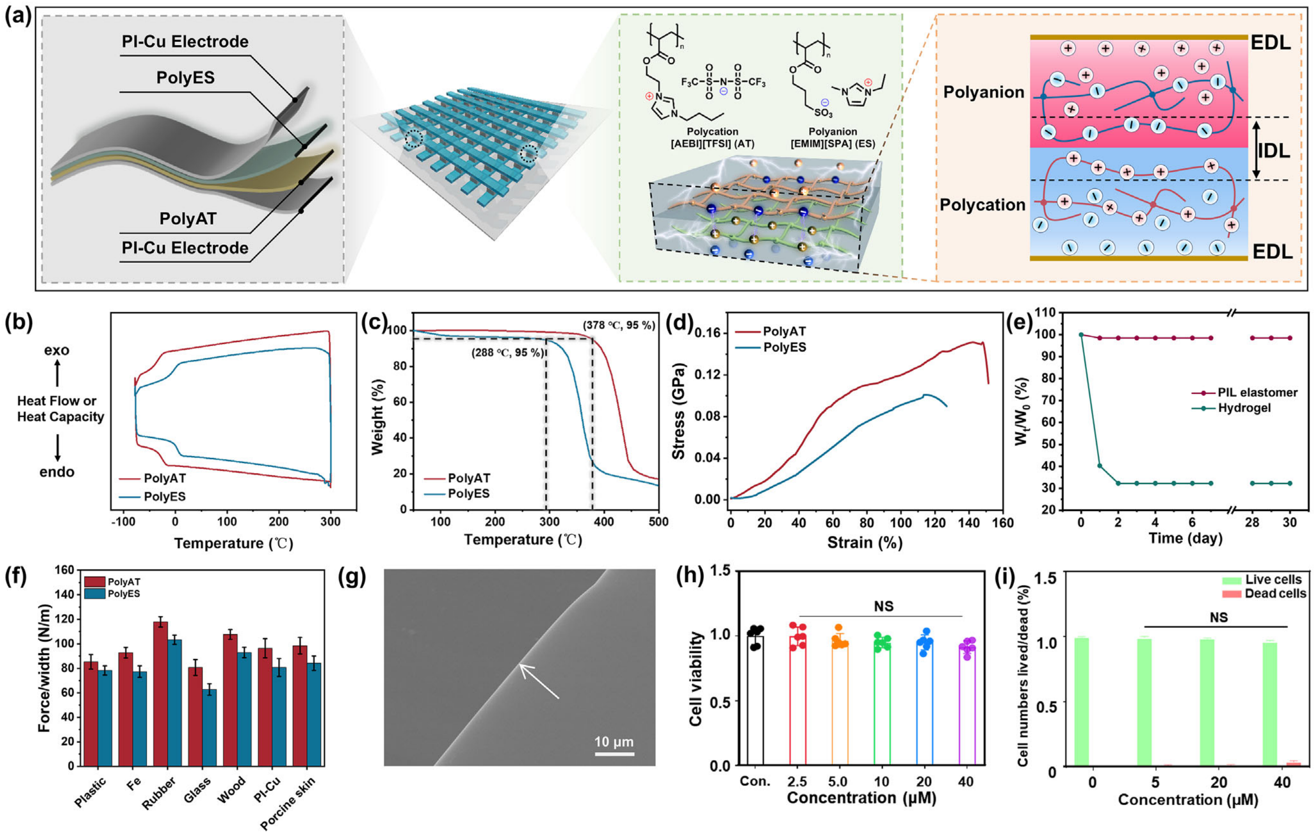

Figure 2. a) Schematic diagram of the PIL elastomer diode-based artificial skin for multimodal perception and memory. b) DSC and c) TGA characterizations of PolyAT and PolyES. d) Stress–strain curves of PolyAT and PolyES. e) Weight loss of hydrogel and PIL elastomer over storage time at room temperature. f) The adhesion strength of PolyAT and PolyES on various substrates quantified through a 90° peel test. g) Interface bonding between PolyAT and PolyES observed through SEM. h) Biocompatibility results of the material verified through the CCK-8 assay. i) Verify the biocompatibility of the material through cell fluorescence imaging analysis.

of the biocompatibility of PolyAT and PolyES using the CCK-8 assay kit. The cell counting results indicated no significant difference between the experimental and control groups ($p > 0.05$), which fully demonstrates that PolyAT and PolyES have good biocompatibility (Figure 2h). The detailed experimental procedures are provided in the Supporting Information.[41,42] To further validate the biocompatibility of PolyAT and PolyES, we conducted an in-depth analysis using cell fluorescence imaging technology. Through intuitive fluorescence imaging and quantitative assessment, the results showed that PolyAT and PolyES did not significantly affect cell viability at different concentrations (Figure 2i; Figure S13, Supporting Information).

2.2. Electrical Perception of the PIL Elastomer Diode-Based Artificial Skin

The PIL elastomer diode is able to respond to current stimulation and outputs a voltage signal that varies with the magnitude and number of input current pulses. This behavior effectively mimics fundamental characteristics of learning and memory in the human brain, such as EPSP, PPF, STP, LTP, STM, LTM, and permanent memory, as illustrated in **Figure** 3a,b.[43–45] Figure 3c,d depicts the output voltage under varying numbers of current pulses at a fixed current intensity, and under different current intensities with a constant number of pulses, respectively. The results indicate that both the output voltage and its retention time increase with the current intensity and pulse number. This behavior resembles the progression from STM to LTM following external stimulation, highlighting that the retention time and memory effects of the device are influenced by both pulse intensity and number (Figure 3e). The possible explanation could be that the electric field drives $EMIM^+$ cations and $TFSI^-$ anions to migrate toward the opposite side, leading to partial recombination of the ions and a reduction in voltage. Once the stimulation ceases, the ions gradually redistribute to their original state, causing the voltage to decay.[46,47] Notably, increasing both the pulse width and intensity extends the retention time of the device's output voltage to over 10^4 s, analogous to permanent memory in the human brain (Figure 3f). When two consecutive electrical pulses (100 nA, 0.22 s) were applied to the PIL elastomer diode, PPF behavior was observed, characterized by an increase in EPSP, as shown in Figure S14 (Supporting Information). The PPF value was calculated using the equation PPF = (A2/A1) × 100%, yielding a value of 139%. The PPF index was further modeled using a bi-exponential function: $PPF = 1 + C1 \cdot \exp(-\Delta T/\tau_1) + C2 \cdot \exp(-\Delta T/\tau_2)$. where C1 and C2 represent the magnitudes of the fast and slow facilitation phases,

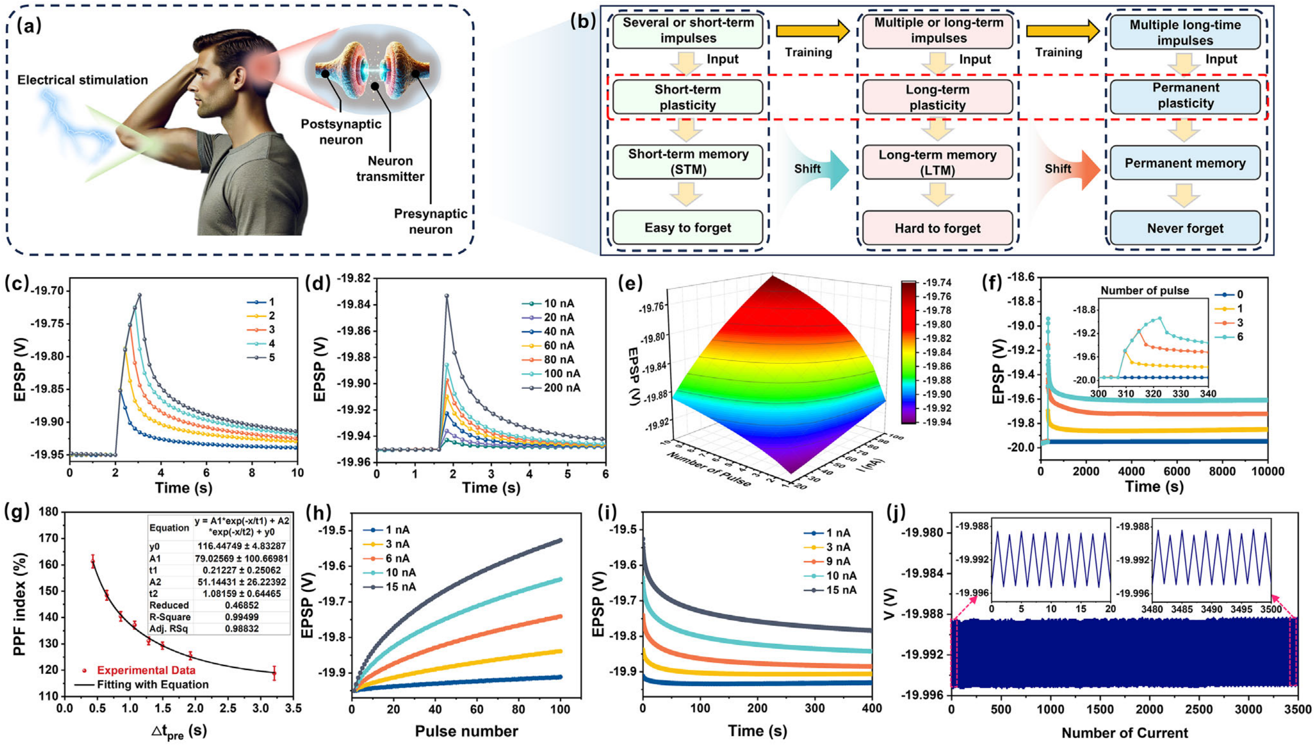

Figure 3. a) Memory behavior of human after electrical stimulation. b) Schematic illustration of the relationship between synaptic plasticity and memory function. The voltage response of the PIL elastomer diode c) under different pulse numbers (current intensity of 150 nA, pulse widths of 0.22 s), d) under different current intensities (pulse widths of 0.2 s), and e) 2D (electrical pulse intensity and number) perception capability of the PIL elastomer diode. f) Under different pulse numbers (current intensity of 200 nA, pulse widths of 2.5 s). g) PPF index as a function of the time interval. h) LTP with the pulse number increased to 100 (pulse widths of 0.2 s). i) Transition from STP to LTP after 100 pulse stimulations at different current intensities. j) Stability over 3500 write-erase cycles.

respectively. Fitting the data revealed $\tau_1 = 212.27$ ms and $\tau_2 = 1081.59$ ms (Figure 3g), values comparable to those observed in biological synapses. In biological systems, the transition from STP to LTP can be achieved through repeated pulse stimulations, leading to a sustained enhancement of synaptic strength. To simulate this transition, we applied 100 identical consecutive pulses to the PIL elastomer diode (Figure 3h). The voltage increased progressively with the number of stimulation pulses, demonstrating an electric induction effect. The retention characteristics of the device were also studied after 100 pulse stimulations, showing a significant increase in retention time over a 400 s observation period (Figure 3i). This confirmed the successful transition from STP to LTP, as evidenced by the enhanced retention time. Furthermore, over 3500 repeated write/erase cycles demonstrated the excellent durability of the PIL elastomer diode (Figure 3j).

2.3. Tactile Sensing of the PIL Elastomer Diode-Based Artificial Skin

The human skin not only exhibits excellent electro-sensing capabilities but also possesses pressure-sensing abilities based on the piezoresistive effect. The human skin tactile system includes both light touch and pressure sensations. Light touch refers to the perception of slight stimulation on the skin's surface, while pressure is the perception of deeper or sustained mechanical forces on the skin.[48,49] Similarly, when pressure is applied to the artificial skin, the ion distribution and the built-in potential across the ionic p-n junction interface would be altered, thereby a highly sensitive tactile perception is expected. As shown in **Figure 4a**, the sensor produced a noticeable and repeatable signal when contacting silk (≈0.075 Pa), demonstrating its stability and accuracy toward ultralow pressure, which is analogous to the human light touch sensation. When a pressure of 1.9 kPa was applied to the sensor, it responded rapidly, and after the object was removed, it showed a clear hysteresis effect, similar to the human pressure sensation (Figure 4b). The mechanism may be that when an external mechanical input is applied, additional ion diffusion occurs, and the volumetric concentration of $EMIM^+$ cations and $TFSI^-$ anions increases with pressure (Figure S15, Supporting Information). This disrupts the original equilibrium state, leading to an expansion of the space charge region (Figure S1, Supporting Information).[50,51]

Figure 4c shows that a single pressure pulse (1.3 kPa, 0.4 s) triggered an EPSC that reached 8.86 μA. After the pressure pulse was removed, the EPSC rapidly decayed and gradually returned to the baseline, which is very similar to the memory behavior of the human brain. By adjusting the pressure intensity, the EPSC value could be easily modulated (Figure 4c). Additionally, by applying two consecutive pressure pulses (1.9 kPa, 0.2 s), PPF

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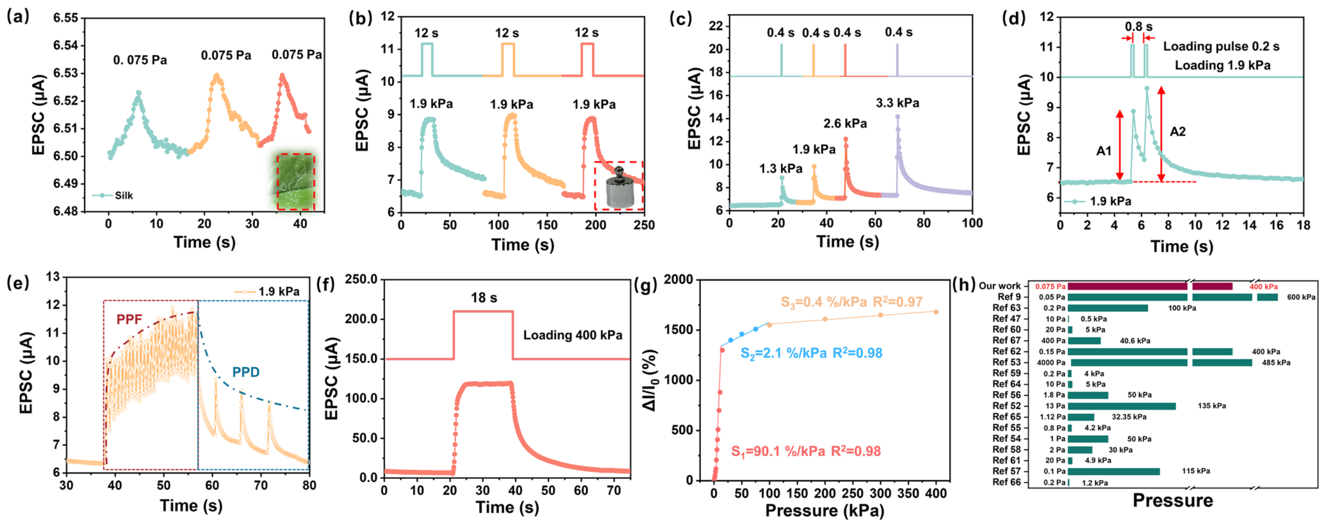

Figure 4. EPSC triggered by a) silk and b) weight. c) EPSC triggered by a variety of loading spikes. d) EPSCs triggered by a pair of loading pulses (1.9 kPa, 0.2 s) with 0.8 s interval. e) EPSCs at a series of consecutive pressure pulses with different interval lengths. f) Response and relaxation time when loading and releasing a pressure of 400 kPa. g) The sensor's sensitivity to pressure and its linear range. h) Comparison of pressure sensing range with previous studies.

behavior was observed, as indicated by the increase in EPSC shown in Figure 4d. According to the equation (PPF = A2/A1 × 100%), the calculated PPF value was 132%. It is noteworthy that this PPF behavior is highly dependent on the interval between pressure pulses, with shorter intervals increasing the EPSC value, while longer intervals eventually lead to a decrease in EPSC. As shown in Figure 4e, a typical PPF was obtained when the interval was 0.5 s, whereas paired-pulse depression (PPD) was observed when the interval increased to 5 s. Furthermore, as shown in Figure 4f, the sensor can accurately sense ultra-high pressure up to 400 kPa. Additionally, fatigue resistance endows the sensor with stable pressure-sensing performance during long-term operation (Figures S16 and S17, Supporting Information). Sensitivity is one of the most important performance indicators of a sensor. As shown in Figure 4g, the sensor exhibits excellent sensitivity characteristics in the 0–400 kPa pressure range. Specifically, in the low-pressure range of 0–30 kPa, its sensitivity reaches as high as 90.1%/kPa. Through comparison with previous similar studies, the detection range of 0.075 Pa to 400 kPa achieved by our sensor is one of the widest detection ranges reported for similar materials (Figure 4h).[9,47,52–67] We conducted a comparative analysis of the pressure sensing performance of the same sensor before and after 30 days of storage. The experimental results show that the device's pressure sensing performance after 30 days of storage remained highly consistent with the initial results, confirming its excellent long-term stability (Figure S18, Supporting Information). In addition, we conducted repeated pressure tests on three sensors, which exhibited high consistency in pressure response, demonstrating the repeatability and reliability of the sensor design (Figure S19, Supporting Information). By comparing the pressure responses before and after humidification, the humidity insensitivity of the sensor was validated (Figure S20, Supporting Information).

2.4. Temperature Perception of the PIL Elastomer Diode-Based Artificial Skin

In addition to excellent tactile functions and memory behavior, this artificial skin can also percept temperature variations. As shown in the **Figure 5a**, when the PIL elastomer diode approaches an object of 0 °C, the current change increases inversely with the distance to the object. Moreover, by applying two consecutive temperature pulses (0 °C, 2.3 s), PPF behavior is observed, as indicated in Figure 5b. According to the equation (PPF = A2/A1 × 100%), the calculated PPF value was 128%. The PPF index can be modeled using a bi-exponential function. Fitting results showed that τ_1 = 2.9 s and τ_2 = 3.6 s (Figure 5c). Further analysis reveals that PPF behavior is highly dependent on the interval between temperature pulses: shorter intervals increase the EPSC value, while longer intervals lead to EPSC decay. As shown in Figure 5d, typical PPF is observed when the pulse interval is 10 s, whereas when the pulse interval increases to 20 s, PPD is observed. Additionally, as illustrated in Figure 5e–g, the sensor can effectively detect a broad range of temperatures from −80 to 120 °C with excellent repeatability in a non-contact sensing mode, well beyond the reach of traditional hydrogel-based artificial skins. This capability not only helps prevent skin burns caused by direct contact with high-temperature sources but also enables a variety of applications, such as non-contact early warning systems. In addition, the sensor exhibits a good linear response to temperature (Figure S21, Supporting Information). We conducted a comparative analysis of the temperature sensing performance of the same sensor before and after 30 days of storage. The experimental results show that after 30 days of storage, the device's temperature sensing performance remained highly consistent with the initial results, confirming the excellent long-term stability of the device (Figure S18, Supporting Information). In addition, we conducted repeated temperature tests on three

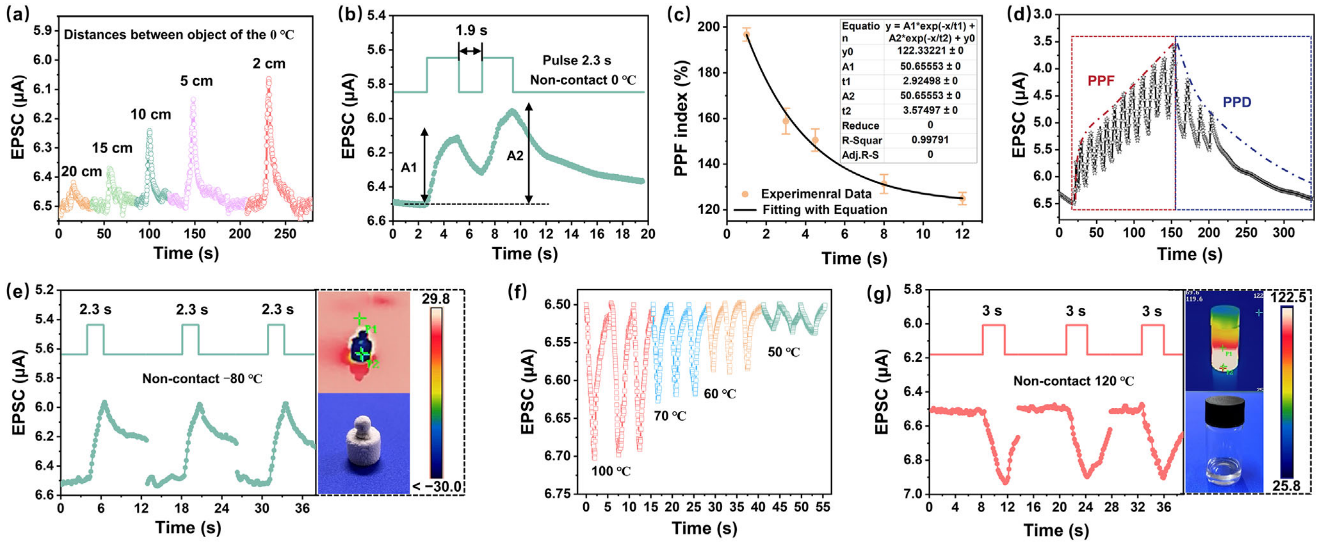

Figure 5. a) EPSC triggered by an object of 0 °C at different distances to the PIL elastomer skin. b) EPSCs triggered by a pair of temperature pulses (0 °C, 2.3 s) with 1.9 s interval. c) PPF index as a function of the time interval. d) EPSCs at a series of consecutive temperature pulses with different interval lengths. EPSC triggered by an object of e) −80 °C, f) 50–100 °C (2 s), and g) 120 °C (the temperature sources are all 5 cm away from the sensor).

sensors. The three sensors showed high consistency in temperature response, demonstrating the repeatability and reliability of the sensor design (Figure S19, Supporting Information). The response mechanism of the PIL elastomer diode to temperature may be that, as temperature changes, ions gain different levels of thermal energy, altering their diffusion rate. Meanwhile, temperature variations affect the mobility of polymer chains and alter the viscosity of the elastomer, thereby modifying ion transport channels.[11,68,69]

2.5. Simultaneous Temperature and Pressure Sensing

In practical applications, real-time monitoring of the spatial distribution of pressure and temperature in a large area is crucial, especially in fields like smart manufacturing, health monitoring, and flexible electronics.[70–72] To address this need, we developed an 8 × 8 sensor array based on PIL elastomer diode, which precisely senses and maps the temperature and pressure distribution of objects by monitoring current variations at different positions. The sensor array is designed so that each PIL elastomer diode module functions as an independent sensing unit. When applying pressure, the unit gives corresponding changes in current, which are directly related to the magnitude of the applied pressure. As shown in **Figure** 6a,b, by analyzing the current changes in each sensor unit, we can successfully locate and identify the positions and the magnitude of the pressure exerted by different objects (such as weight and blade).

When the sensor array approaches objects with varying temperatures, the current changes in the sensing units nearest to the heat source become more pronounced. By monitoring the degree of current changes in each pixel, it is possible to accurately determine the position of the objects and their distance from the sensor. This capability enables the sensor array to detect not only pressure but also spatial temperature distribution. As shown in Figure 6c,d, when an object (such as a 50 °C steel ruler or a 50 °C glass bottle filled with silicone oil) approaches the sensor array, multiple sensing units generate different currents. The closer to the heat source, the more significant the current changes, and these changes are accurately recorded through multi-point feedback from the array, reflecting the object's position and temperature. The 8 × 8 sensor array based on PIL elastomer diode exhibits exceptional multifunctionality, enabling real-time, efficient sensing and precise mapping of the pressure and temperature distribution of objects.

The development of intelligence is a key trend in the evolution of robotics, with sensor technology playing a crucial role in this process. Although current robotic hands have already achieved basic human hand functions, such as picking up fragile items like eggs[73] and tomatoes,[74] the full integration of temperature and pressure sensors into robotic hand remains largely underutilized. To demonstrate the potential of this PIL elastomer diode sensing system in practical applications, we attached the PIL elastomer diode to a robotic hand (Figure 6i). When humans pick up objects, their skin detects the temperature. If the temperature is too high or too low, the skin automatically provides immediate feedback to avoid direct contact, guiding individuals to choose objects with suitable temperatures. This natural reflex prevents burns or frostbite and aids in decision making. Similarly, we attached the sensor to the robotic hand, enabling it to function as human skin. With this sensor, the robotic hand can detect both the temperature and pressure of objects, effectively mimicking the human process of selecting appropriate objects (Figure 6e).

In the experiment, the robotic hand was tasked with selecting from nine objects of unknown temperature and mass. As shown in Figure 6f,g, the objects had three different temperatures (−80, 120, 25 °C), with each temperature corresponding to three different masses (5, 8, 13 g). By analyzing the temperature and mass data of the objects (Figure 6h), the robotic hand was able to intelligently select the target object with the desired temperature and mass (Figure 6i). This process not only endows the robotic

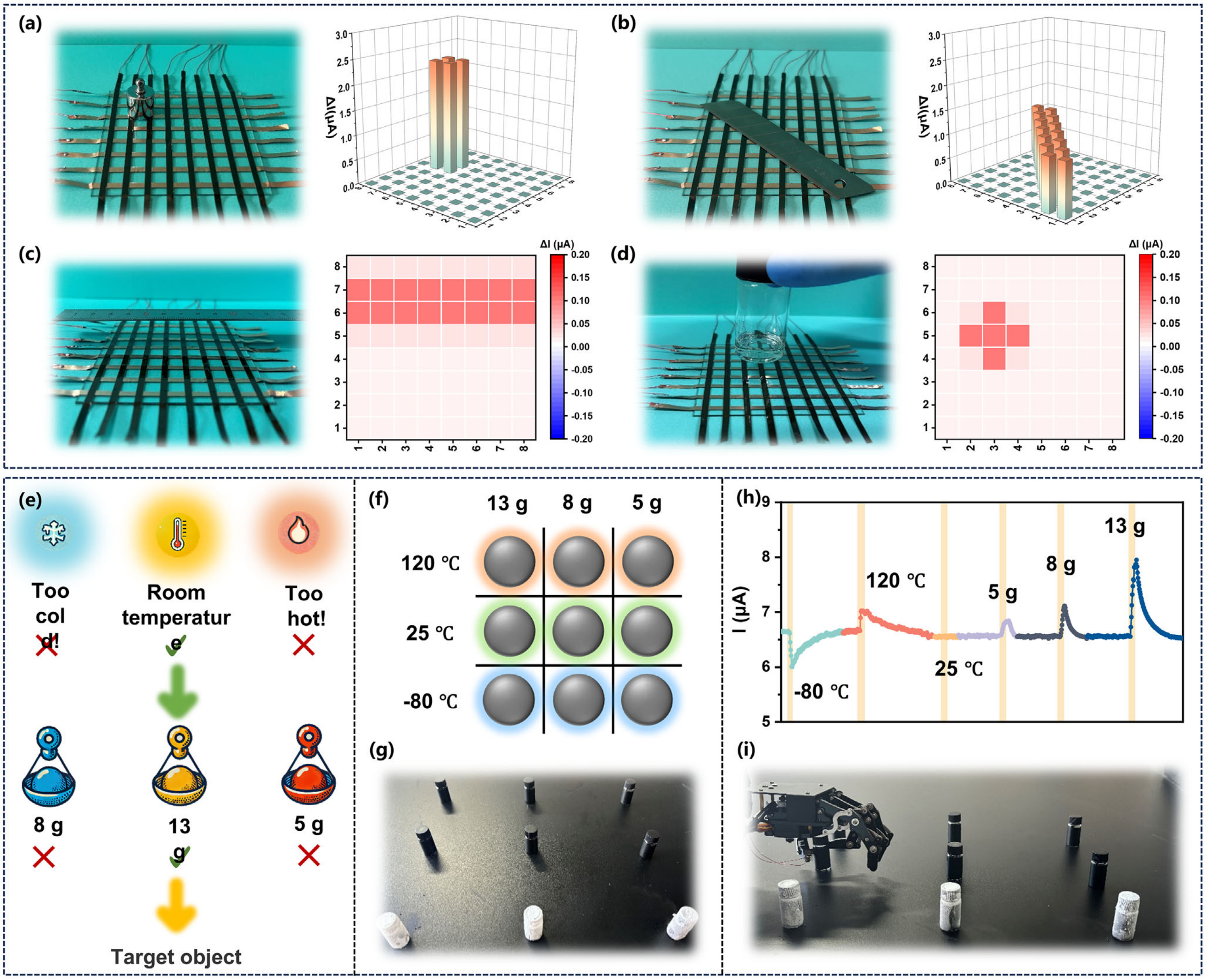

Figure 6. The response of an 8 × 8 sensor array to a) a weight and b) blade. The response of an 8 × 8 sensor array to c) a steel ruler of 50 °C and d) a glass bottle filled with silicone oil of 50 °C. e) Schematic diagram of robot arm grasping the target with specified temperature and weight. f) Schematic diagrams and g) photographs of objects with different temperatures and weight. i) The robot arm takes the object according to h) the response curve (the temperature sources are all 5 cm away from the sensor).

hand with intelligent perception and decision-making but also highlights its improved efficiency in handling complex tasks.

2.6. In-Sensor RC for MNIST Handwritten Images Recognition

By combining LTM and STM memory behaviors triggered by electrical pulses, we propose a fully memristive diode based in-sensor RC. This system can directly learn and infer from electrical signals without the need for dedicated electrical sensors or digital computers.[75,76] First, PIL elastomer diode can serve as the memory unit in the readout layer of the RC system. **Figure 7a** shows multi-bit potentiation and depression states triggered by two pulse sequences in the PIL elastomer diode. Upon applying 50 positive electrical pulses (current intensity of 15 nA, pulse width: 0.2 s), multi-level potentiation states are observed. Conversely, multi-level depression states are observed when 50 negative electrical pulses (current intensity of −9.5 nA, pulse width: 0.2 s) are applied. These multi-bit states, controlled by the number of pulses, serve as the foundation for the real-time learning capabilities of the in-sensor RC system utilizing the PIL elastomer diode. The nonlinear dynamic changes in channel voltage induced by electrical stimulation in the PIL elastomer diode are key to achieving RC. These dynamics, characterized by STM, are demonstrated by applying various electrical pulse sequences to the PIL elastomer diode (Figure 7b). Each pulse sequence consists of four pulses, where "1" and "0" represent electrical pulses with currents of 150 nA and 0, respectively, with a pulse width of 0.21 s. As shown in Figure 7c,d, 16 electrical pulse sequences ranging from (0000) to (1111) generate 16 clearly distinguishable states, highlighting the powerful capability to map complex spatiotemporal signals into storage states.

In our simulation experiments, for simplicity, all optical images are set to a resolution of 28 × 28 pixels (28 pixels in both

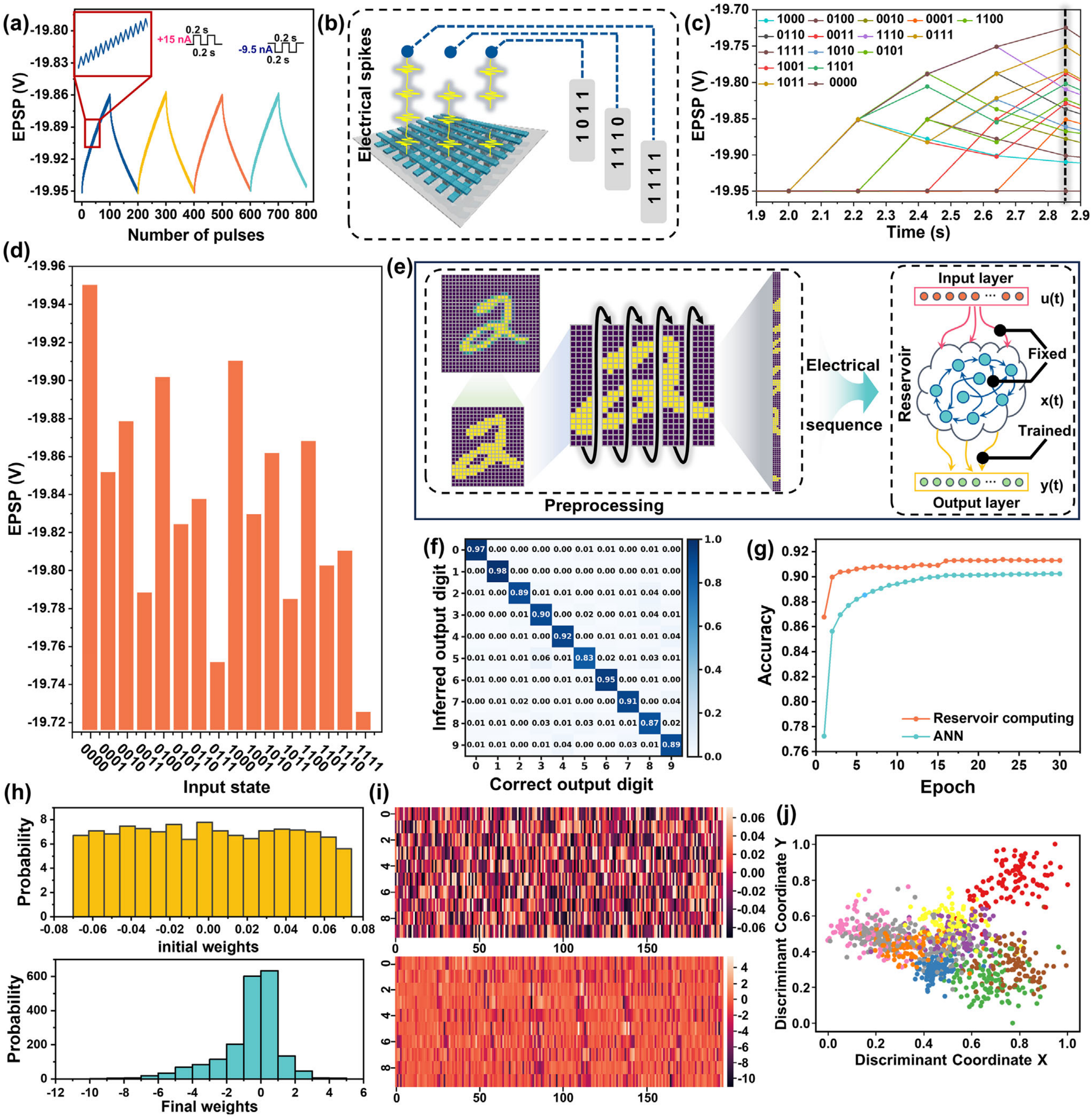

Figure 7. a) Multi-bit conductance evolution of the PIL elastomer diode triggered by 100 pulse sequences. b) Illustration of electrical pulse trains applied to the PIL elastomer diode. c) Electrical-response to 16 different pulse streams of PIL elastomer diode-based RC. (electrical pulse frequency: 1.17 Hz). d) Initial and final voltage values of the reservoir states under 16 combinations of pulse stimuli. e) Schematic illustration of the RC system for classifying the MNIST dataset. f) Confusion matrices of the PIL elastomer diode RC system on recognizing MNIST. g) The recognition results. The cyan and orange curves correspond to the classification accuracy of the software and PIL elastomer diode based RC system, respectively. h) Initial (top panel)/final (bottom panel) corresponding PIL elastomer diode array conductance distributions of i) before/after training. j) Dimensionality reduction of the reservoir outputs using LDA.

vertical and horizontal directions) As shown in Figure 7e, these images are first binarized and cropped to remove blank edge pixels. The remaining pixels are reshaped into electrical pulse sequences with a width of 4 and a height of 110, which are then fed into a dynamic reservoir of 110 visual neurons made from PIL elastomer diode. Each row of the matrix, represented as a 4-bit row vector, corresponds within each category. The training results show a simulated classification accuracy of to one of the 16 electrical stimulation patterns. As depicted in Figure 7f, the confusion matrix for handwritten digit classification demonstrates that most elements are concentrated near the diagonal, indicating strong system robustness. The training results are summarized in Figure 7g, where the cyan dot represents the generational classification accuracy in software (90.2%), and the orange dot represents the in-sensor RC results using the PIL elastomer diode (91.3%), which slightly surpasses the software benchmark. The initial and final conductance values of the electrically operated memory devices in the readout layer, along with their distributions, are presented in Figure 7h,i, respectively. Figure 7j illustrates the 2D distribution of feature vectors extracted by the reservoir, with dimensionality reduced using Linear Discriminant Analysis (LDA). The results demonstrate that due to the nonlinear transformation of STM in the electrical operation storage devices, the encoded features are well-clustered in the 2D space.

3. Conclusion

In this study, we have successfully developed a bioinspired ionic skin, inspired by the electrocyte of electric eels, and achieved a flexible, transparent, and biocompatible sensor capable of transmitting ionic signals. This multifunctional iontronic skin exhibits excellent tactile, thermal, and electrical sensing capabilities and memory, enabling the detection of a wide range of pressures (0.075 Pa to 400 kPa), temperatures (−80 to 120 °C), and currents (1to–200 nA). Moreover, the sensor demonstrates rich synaptic behaviors, allowing for effective sensing and memory toward multimodal external stimuli. Leveraging these behaviors, this PIL elastomer diode based artificial skin is capable of recognizing handwritten digits with an accuracy of 91.3% when coupled with in-sensor RC. Moreover, integrating the ionic skin on a robotic hand enables to precisely grasp objects of temperatures and weights on demand, further showcasing its potential in wearable electronics, soft robotics, and smart prosthetics. Compared to traditional hydrogel-based sensors, this material offers superior stretchability, wider electrochemical window, higher ionic conductivity, and resistance to water evaporation and freezing, making it an ideal candidate for next-generation artificial skins. This work marks a significant step forward in bridging the gap between biological and artificial sensory systems, providing a solid foundation for further research in fields such as human-machine interaction, augmented reality, and advanced sensing technologies.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords

in-sensor reservoir computing, iontronic artificial skin, multimodal perception, p-n junctions, rich synaptic behaviors

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